



High ionic conductivity of new polymer electrolytes based on high molecular weight polyether comb polymers

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Abstract—High molecular weight polyether comb polymers, poly[ethylene oxide-*co*-2-(2-methoxyethoxy)ethyl glycidyl ether] P(EO/MEEGE), have been used as matrixes of polymer electrolytes. The high molecular weight ($> 10^6$) polymers give self-standing and elastic polymer electrolyte films without chemically cross-linked structures at room temperature, when complexed with electrolyte salts. With increasing the composition of MEEGE in the copolymers, the degree of crystallinity of the polymer electrolytes decreases, and the ionic conductivity appreciably increases. The increase in the conductivity cannot be explained only by the crystallinity decrease, the increase in the number of highly mobile ether side chains with the increase in MEEGE composition contributes to achieving the high conductivity. The fast ion transport would be realized in cooperation with the fast side chain motion. The polymer electrolytes complexed with lithium bis(trifluoromethylsulfonyl)imide (LiTFSI) exhibit higher conductivity than those complexed with a conventional salt, lithium perchlorate. The polymer electrolytes, P(EO/MEEGE) containing 9 mol% of MEEGE complexed with LiTFSI, exhibit high ionic conductivities of 10^{-4} S cm⁻¹ at 30°C and 10^{-3} S cm⁻¹ at 80°C. An electrochemically stable potential window, established by solid state cyclic voltammetry using microelectrodes, is at least 4 V from the Li/Li⁺ potential. © 1998 Published by Elsevier Science Ltd. All rights reserved

Key words: polymer electrolyte, ionic conductivity, comb polymer, lithium bis(trifluoromethylsulfonyl)imide, microelectrode.

INTRODUCTION

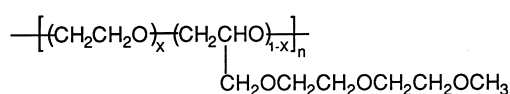
Polymer electrolytes have highly potential applicability as solid electrolytes to advanced electrochemical devices such as high energy density solid batteries and electrochromic windows [1–3]. Global interest has especially focused on lithium/polymer electrolyte batteries as energy sources of electric vehicles, because of their high energy density, safety, and the flexibility in both their shapes and production processes [1–3]. However, the ionic conductivity of conventional poly(ethylene oxide) based polymer electrolytes has not been of a satisfactory level for the applications, especially the ionic con-

ductivity at ambient temperatures is much lower than that of liquid electrolyte solutions.

Many studies to overcome the low ionic conductivity of polymer electrolytes have been made [1–3]. To attain highly conducting polymer electrolytes, we have developed network polymer electrolytes having short and flexible ether side chains. A new monomer to synthesize the network polymer electrolytes is 2-(2-methoxyethoxy)ethyl glycidyl ether (MEEGE) [4–7]. MEEGE can be homopolymerized and copolymerized with ethylene oxide (EO) by anionic ring-opening polymerization using base catalysts to yield polyether oligomers [4,5]. These oligomers can be converted to macromonomers, so-called macromers, by introducing acrylate functionality to terminal hydroxyl groups of the oligomer

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chain ends. The network polymer electrolytes have been prepared by radical polymerization of the macromonomers having on the average more than one acrylate functionality [5, 6]. The preparation of network polymer electrolytes having flexible and ion-coordinating ether side chains to achieve high ionic conductivity is based on the concept of the coupling of fast ionic transport and fast side chain motion. The PMEEGE-based network polymer electrolytes have achieved comparably high ionic conductivity and the effect of the introduction of flexible ether side chains on the ionic conductivity have been proved [6, 7].



Scheme 1. Structure of P(EO/MEEGE).

In this study, high molecular weight ($>10^6$) comb-type copolymers consisting of EO and MEEGE [P(EO/MEEGE)] are used as host polymers of polymer electrolytes (Scheme 1). The high molecular weight might allow self-standing polymer electrolyte films without chemically cross-linked structures. High ionic conductivity, as seen in the network polymer electrolytes based on P(EO/MEEGE) macromonomers, is also expected due to the same polymer structure. The influence of the introduction of flexible ether side chains on the polymer electrolytes in terms of ionic conductivity has been investigated by comparing the ionic conductivity data for the P(EO/MEEGE) polymer electrolytes having different copolymer compositions. Lithium bis(trifluoromethylsulfonyl)imide (LiTFSI), which generally gives higher ionic conductivity when complexed with polyethers than conventional salt [8, 9], is used as a lithium salt. A comparison of thermal and ionic conductivity data, and lithium transference numbers for the polymer electrolytes complexed with LiTFSI and conventional lithium salt LiClO₄ has been carried out. The electrochemically stable potential window is also presented.

Table 1.
Characteristics of high molecular weight P(EO/MEEGE)

Abbreviation	Composition ^a ([EO]:[MEEGE])	Molecular weight ^b	M_w/M_n^c	T_g^d (°C)	T_m^d (°C)	ΔH_m^d (J/g)	Degree of crystallinity ^d (%)
P(EO/MEEGE)-5	95:5	$>4.0 \times 10^6$ ^c	—	-61	47	78	38
P(EO/MEEGE)-9	91:9	1.5×10^6	5.4	-65	39	28	14

^aDetermined by ¹H-NMR spectra.

^bWeight-average molecular weight determined by gel permeation chromatography (GPC).

^cPolydispersity index, where M_w and M_n are weight average and number average molecular weight.

^dDetermined by DSC. Degree of crystallinity was evaluated by using ΔH_m and the heat of fusion of 100% crystalline PEO (203 J/g).

^eHigher than exclusion limit molecular weight of GPC column.

EXPERIMENTAL

Preparation of polymer electrolytes

High molecular weight P(EO/MEEGE) copolymers were kindly supplied by Daiso Co., Ltd. The molecular characteristics and the abbreviation of the copolymers are shown in Table 1. Gel permeation chromatography (GPC) was performed by using 0.01 M LiBr *N,N*-dimethylformamide as the elution solvent at 60°C, and the weight-average molecular weight (M_w) and number-average molecular weight (M_n) were calculated based on the calibration of columns using poly(ethylene oxide) standards. The copolymers were dried at 50°C under high vacuum and stored in an argon atmosphere glove box (VAC , $[\text{O}_2] < 1$ ppm, $[\text{H}_2\text{O}] < 1$ ppm). Reagent grade LiClO₄, dried under high vacuum at 140°C for 12 h, and LiTFSI dried under high vacuum at 180°C for 12 h were kept in the glove box and used as electrolyte salts.

In the argon atmosphere glove box, polymer electrolytes were prepared from the P(EO/MEEGE) copolymers and the anhydrous salts. Stoichiometric amounts of LiClO₄ or LiTFSI and a P(EO/MEEGE) copolymer were dissolved in anhydrous acetonitrile to form a highly viscous homogeneous solution. The viscous solution was cast on a polytetrafluoroethylene (PTFE) plate. In order to obtain homogeneous and flat films, the solvent was allowed to evaporate at room temperature in the glove box for several hours and then under high vacuum for 12 h.

Measurements

Differential scanning calorimetry (DSC) of P(EO/MEEGE) and the polymer electrolytes was performed on a Seiko Instruments DSC 220C in nitrogen atmosphere. The samples sealed in Al pans in the dry glove box were heated up to 100°C and quenched to -120°C and then heated at a rate of 10 K/min up to 80°C. The DSC thermograms were recorded during the programmed heating cycle.

Dynamic mechanical measurements of P(EO/MEEGE) films were conducted by using a Seiko Instruments DMS 210 in nitrogen atmosphere at a heating rate of 1 K/min.

Ionic conductivity was determined by means of the complex impedance measurements with stainless-steel blocking electrodes, using a computer controlled Hewlett-Packard 4192A LF impedance analyzer over the frequency range from 5 Hz to 13 MHz. The polymer electrolyte films (0.2 mm in thickness) were cut into disks of 10 mm in diameter, and put into PTFE ring spacers (13 mm in outer diameter, 10 mm in inner diameter, 0.2 mm in thickness). The polymer electrolytes films with PTFE ring spacers, sandwiched between stainless-steel electrodes, were sealed in PTFE containers and were subjected to the complex impedance measurements.

Apparent transference number of Li^+ ion in the polymer electrolytes was determined by means of the combination of complex impedance measurements and potentiostatic *dc* polarization measurements using metallic lithium electrodes [10, 11]. The amplitude of complex impedance measurements and the *dc* polarization voltage was 10 mV.

Electrochemically stable potential window of the polymer electrolytes was checked by solid state cyclic voltammetry using a three-electrode microcell described in a previous study [12]. The three-electrode microcell consists of tips of three electrodes: Pt microelectrode (10 μm diameter, sealed in soft glass capillary), and Pt and Ag wires counter and pseudo-reference electrodes, respectively, all sealed in an insulating epoxy cylinder. A viscous acetonitrile solution of P(EO/MEEGE) and LiTFSI or LiClO_4 was cast on the three-electrode cell in the argon atmosphere glove box, and the solvent was allowed to evaporate under high vacuum. All the cyclic voltammetry measurements were performed in argon atmosphere at 60°C.

RESULTS AND DISCUSSION

Characterization of polymer electrolytes

The P(EO/MEEGE) copolymers have fairly high molecular weight (Table 1). When a base catalyst, such as KOH, is used for the ring-opening copolymerization of EO and MEEGE, the maximum molecular weight of the resulting polymers is several thousands [5, 6]. Furthermore, it is well known that the random copolymerization of EO with other epoxides is usually difficult, because of relatively higher reactivity of EO than the other epoxides. For the copolymerization of the present P(EO/MEEGE), a condensation product of an organic tin compound, typically dibutyltin oxide, and a phosphoric acid ester, typically tributylphosphate, was used as a highly active catalyst for the ring-opening polymerization [13]. Even by using the highly active catalyst, the polymerization rates and molecular weights of the resulting polymers decreased with increasing the molar ratio of MEEGE in feed. Thus, in the actual polymerization reactions,

MEEGE was initially introduced into a reaction vessel at one time, whereas EO was fed successively to the vessel with gas-chromatographically monitoring the consumption of MEEGE and EO at a predetermined composition of the resulting copolymer. Molecular weights of the resulting P(EO/MEEGE)-5 and P(EO/MEEGE)-9 were higher than 10^6 . Although the molecular weight distribution was rather large (Table 1), the GPC chromatograms were unimodal (not shown) with no evidence of independent polymerizations of EO rich high polymers and MEEGE rich oligomers. The lowest molecular weight eluted from the GPC columns was 6000 in the case of P(EO/MEEGE)-9. The degree of crystallinity of P(EO/MEEGE)-5 and P(EO/MEEGE)-9 is 38 and 14%, respectively (Table 1), which is considerably lower than that of high molecular weight poly(ethylene oxide) (normally higher than 70%). Considering these facts, it is concluded that EO and MEEGE can be copolymerized by using the highly active catalyst to give high molecular weight P(EO/MEEGE) copolymers. Reflecting the high molecular weights, the films of the pure polymers and the polymer electrolytes are self-standing elastomers without cross-linked structures and have favorable mechanical strength at room temperature.

Figure 1 shows the tensile modulus (E'), loss modulus (E''), and loss tangent ($\tan\delta$) of P(EO/MEEGE)-5 at 10 Hz as a function of temperature. The main relaxation in E' and the main peaks seen in E'' and $\tan\delta$ at *ca.* -60°C can be assigned to glass transition temperature (T_g) of the polymer. At lower temperature than the main relaxation temperature, another sub-relaxation is seen (*ca.* -120°C). This might correspond to local mode, possibly side chain, relaxation. The E' value at 30°C is still of the order of 10^7 Pa due to the presence of crystallinity, and the mechanical relaxation reaching to flow region ($E' \cong 10^5$ Pa) is followed at *ca.* 50°C , due to melting of the crystallites (Table 1).

The DSC thermograms of P(EO/MEEGE)-9 and its polymer electrolytes complexed with LiTFSI are shown in Fig. 2. P(EO/MEEGE)-9 itself is a partially crystalline polymer having melting point (T_m) of 39°C (endothermic peak temperature) and T_g of -65°C . P(EO/MEEGE)-5 gave T_m of 47°C and T_g of -61°C (Table 1). The degree of crystallinity is a little higher for P(EO/MEEGE)-5. The crystallinity in both of the P(EO/MEEGE) copolymers is based on the crystallization of PEO sequence in the copolymers, and the copolymerization of MEEGE with EO effectively reduces the crystallinity of the resulting P(EO/MEEGE) copolymers (*vide supra*). With increasing the concentration of LiTFSI or LiClO_4 to form the polymer electrolytes, the crystallinity is further reduced. In the case of P(EO/MEEGE)-9 complexed with LiTFSI, the crystallinity is observed till the concentration of LiTFSI reaches $[\text{Li}]/[\text{O}] = 0.04$, which is the ratio of Li atoms to all the

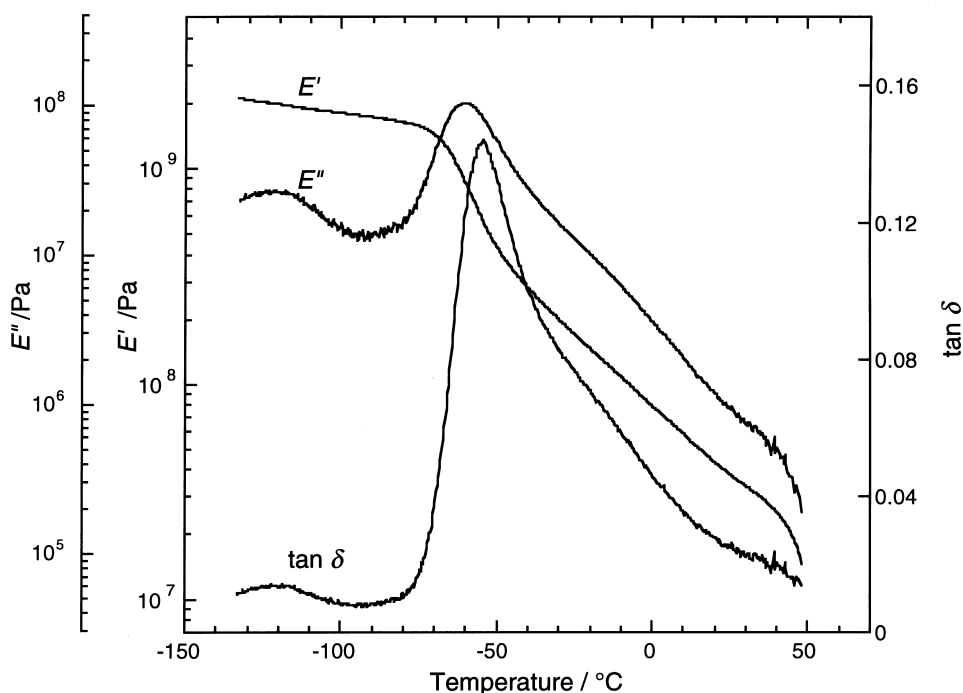


Fig. 1. Tensile modulus (E'), loss modulus (E''), and loss tangent ($\tan\delta$) of P(EO/MEEGE)-5 at 10 Hz as a function of temperature.

ether oxygen atoms in both EO and MEEGE structures. When $[\text{Li}]/[\text{O}]$ is higher than 0.06, the polymer electrolytes become completely amorphous, as shown in Fig. 2. The same results were observed when P(EO/MEEGE)-9 was complexed with LiClO_4 . On the other hand, when P(EO/MEEGE)-5 were used as a matrix polymer to form the polymer electrolytes, the crystallinity persisted up to the salt concentration of $[\text{Li}]/[\text{O}] = 0.06$ for both of LiTFSI and LiClO_4 , and the completely amorphous polymer electrolytes were obtained only at $[\text{Li}]/[\text{O}] = 0.08$.

Figure 3 shows T_g of the polymer electrolytes determined by DSC plotted against either LiTFSI or LiClO_4 concentration. In both of the copolymers, the increase in T_g with the salt concentration is observed, as has been reported in many polyether-based polymer electrolytes [1,2]. Another characteristic is that the increase in T_g with salt concentration is much more pronounced in the LiClO_4 doped polymer electrolytes than in the LiTFSI doped ones. This behavior has also been reported and explained by the striking characteristic of the TFSI anion; the TFSI anion has much greater opportunities for charge delocalization, which promotes dissociation and weakens the interaction with polymer chains [8,9]. The charge delocalization has been ascribed to the strong electron-withdrawing effect of the $-\text{SO}_2\text{CF}_3$ group [8,9]. We can also notice the nonlinear increase of T_g with the salt concentration, especially at low salt concentration. This is well contrasted with the

linear T_g increase in the P(EO/MEEGE) network polymer electrolytes doped with LiTFSI or LiClO_4 , as reported before [6]. The network polymer electrolytes were completely amorphous irrespective of the salt concentration, and T_g of the salt-free network was *ca.* 200 K. On the other hand, the present polymer electrolytes are partially crystalline up to $[\text{Li}]/[\text{O}] = 0.04$ and 0.06 for P(EO/MEEGE)-9 and P(EO/MEEGE)-5, respectively, and T_g of the salt-free polymers is *ca.* 210 K. The higher T_g of the present salt-free copolymers can be assigned to the presence of crystallinity. The crystallites function as physical crosslinking points of the polymers. Furthermore, if the salt is assumed to be incorporated in the amorphous phase in the partially crystalline polymer electrolytes, the real concentration in the amorphous phase should be higher than the apparent concentration. It is considered that the presence of crystalline phase and the resulting condensed effect of salt concentration in the amorphous phase are responsible to the nonlinear increase in T_g in the present polymer electrolytes at the low salt concentrations.

Ionic conductivity

Although the polymer electrolyte films have good mechanical property at room temperature, the creep phenomena were more pronounced with increasing temperature. In order to avoid dimensional change of the polymer electrolytes, PTFE ring spacers were used for the conductivity measurements. Figures 4–6 show the temperature dependences of ionic

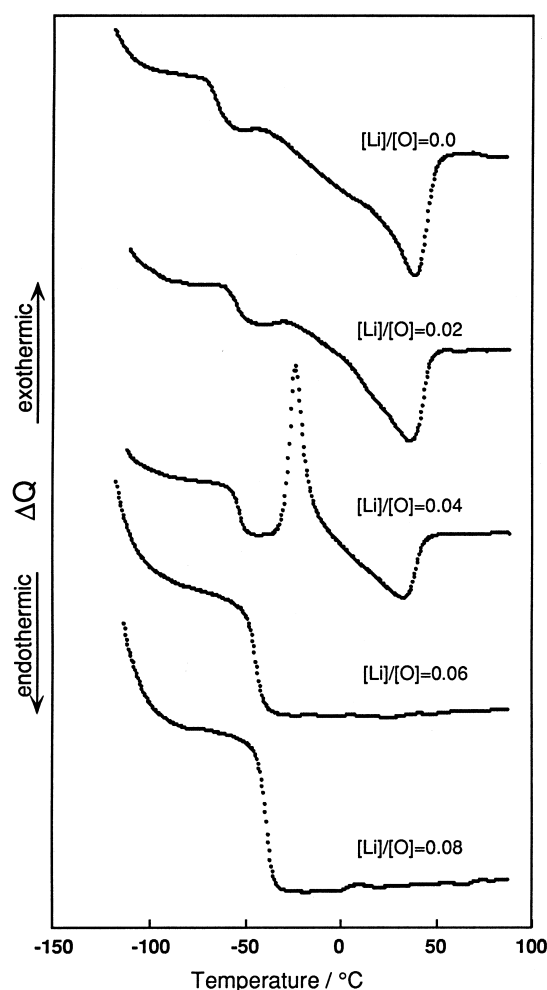


Fig. 2. DSC thermograms of P(EO/MEEGE)-9 and its polymer electrolytes complexed with LiTFSI at a heating rate of 10 K/min after quenching from 100°C.

conductivity for P(EO/MEEGE)-9/LiTFSI, P(EO/MEEGE)-9/LiClO₄, and P(EO/MEEGE)-5/LiTFSI polymer electrolytes, respectively. All of the Arrhenius plots of the ionic conductivity exhibit positively curved-profiles, as can be expressed by WLF or VTF equations. In some polymer electrolytes, the temperature dependences are deviated from smoothly-changed curved-profile and the steep drop of the ionic conductivity is observed below 25°C. These deviation can be attributed to the crystallization of the polymer electrolytes and well coincides with the DSC results. It should be noted that the present polymer electrolytes exhibit appreciably high ionic conductivity at room temperature, especially when P(EO/MEEGE)-9 is used as the matrix. The ionic conductivity data for the polymer electrolytes in this study at 80°C and 30°C are summarized in Table 2. We can notice two remarkable points in this table. First, it is evident that the polymer electrolytes complexed with LiTFSI exhibit higher ionic conductivity than those

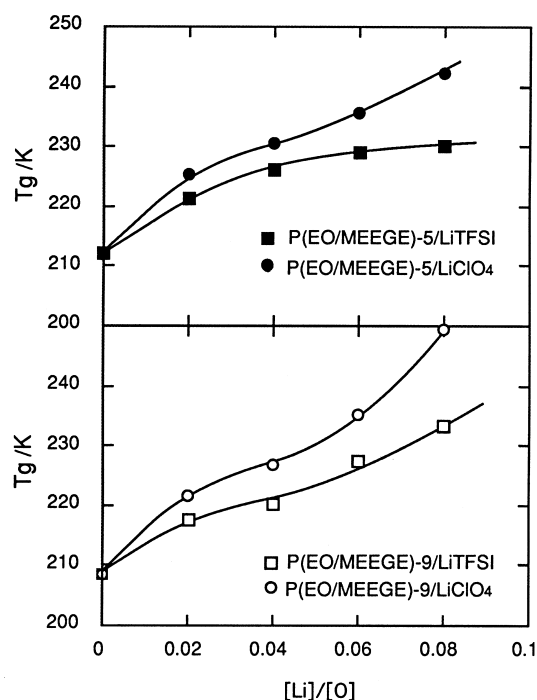


Fig. 3. Glass transition temperature (T_g) of P(EO/MEEGE) copolymers and their polymer electrolytes plotted against salt concentration ($[Li]/[O]$).

complexed with LiClO₄. These differences of ionic conductivity depending on lithium salt species can be attributed to the difference in T_g depending on the lithium salt species, as shown in Fig. 3. Secondly, the ionic conductivity of P(EO/MEEGE)-9 electrolytes is twice as high as that of P(EO/MEEGE)-5 electrolytes. The difference indicates that the copolymerization of MEEGE with EO is effective in terms of the ionic conductivity. One of

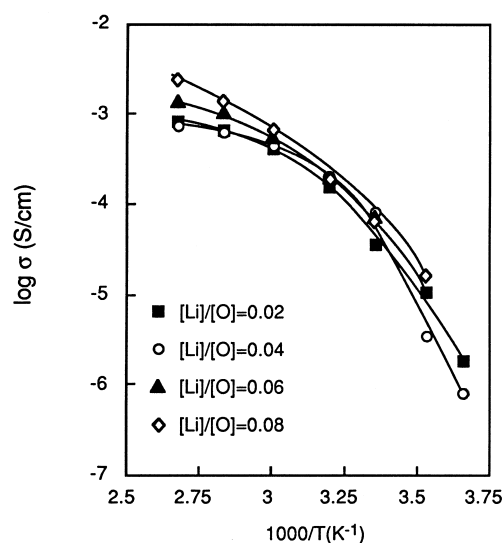


Fig. 4. Arrhenius plots of ionic conductivity for P(EO/MEEGE)-9/LiTFSI electrolytes.

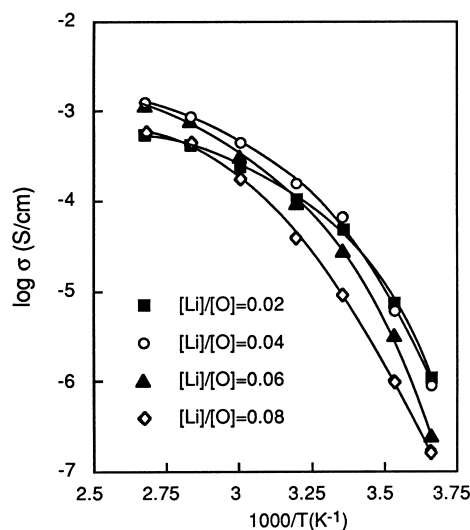


Fig. 5. Arrhenius plots of ionic conductivity for P(EO/MEEGE)-9/LiClO₄ electrolytes.

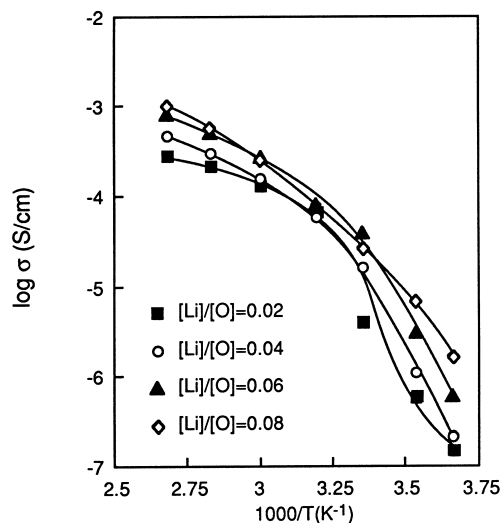


Fig. 6. Arrhenius plots of ionic conductivity for P(EO/MEEGE)-5/LiTFSI electrolytes.

these effects can be interpreted by lowering the crystallinity of the polymer electrolytes, with introducing MEEGE into the polymer backbone. However, it can also be noticed that the conductivity of P(EO/MEEGE)-9 based polymer electrolytes is higher than that P(EO/MEEGE)-5 based ones even if the comparison is made in the completely amorphous electrolytes, as seen in the $[Li]/[O] = 0.08$ electrolytes in Figs 4 and 6. It is also interesting to note that T_g of the P(EO/MEEGE)-9 electrolyte is a little bit higher than that of the P(EO/MEEGE)-5 electrolyte, when both are complexed with LiTFSI at $[Li]/[O] = 0.08$. MEEGE introduces many flexible side chains in the copolymers. The side chains may be highly mobile and the mobility should be much faster than that of the main chain. The ionic conductivity of P(EO/MEEGE)-9/LiTFSI electrolytes reaches $10^{-4} \text{ S cm}^{-1}$ at 30°C and $10^{-3} \text{ S cm}^{-1}$ at 80°C , as shown in Table 2. The conductivity is the highest value ever reported in the solvent-free polyether-based polymer electrolytes, as far as the authors have noticed. There remains one thing which might be taken into account as a reason for

the high ionic conductivity. The molecular weight distribution of P(EO/MEEGE)-9 is rather wide (Table 1), and the lowest molecular weight limit in the copolymer is 6000 from the GPC analysis (*vide supra*). The low molecular weight fraction might function as plasticizers in the high molecular weight matrix. However, we deny this possibility, for the following reasons. The ion transport in polymer electrolytes occurs cooperatively with the local segmental motion, which is generally scaled by T_g of the host polymers. As shown in Fig. 3, T_g 's of the present polymer electrolytes are not specially low, compared with those of the conventional polyether based electrolytes. Furthermore, it has been clarified [14] that the molecular weight has no significant effect on cation mobility above a critical molecular weight of 3200 in poly(ethylene oxide). Although the present copolymers are not poly(ethylene oxide) but its copolymers, we consider that the copolymers also have a similar critical molecular weight, being lower than 6000. At the present, we believe that the fast ion transport in the P(EO/MEEGE) copolymers is cooperated by the fast side

Table 2.
Ionic conductivity of P(EO/MEEGE) complexed with LiClO₄ or LiTFSI

Polymer	$[Li]/[O]$	Salt	Ionic conductivity (S/cm)	
			at 80°C	at 30°C
P(EO/MEEGE)-5	0.06	LiClO ₄	3.3×10^{-4}	2.5×10^{-5}
		LiTFSI	5.1×10^{-4}	5.8×10^{-5}
	0.08	LiClO ₄	2.9×10^{-4}	1.9×10^{-5}
		LiTFSI	5.6×10^{-4}	5.3×10^{-5}
P(EO/MEEGE)-9	0.06	LiClO ₄	7.5×10^{-4}	5.9×10^{-5}
		LiTFSI	1.0×10^{-3}	9.2×10^{-5}
	0.08	LiClO ₄	7.5×10^{-4}	2.4×10^{-5}
		LiTFSI	1.4×10^{-3}	1.3×10^{-4}

chain motion, which should be clarified in the near future.

Apparent Li^+ transference number and electrochemical stability

Apparent Li^+ transference number for the polymer electrolytes, evaluated by the combination of complex impedance and potentiostatic *dc* polarization measurements, are summarized in Table 3. Before the *dc* polarization, the complex impedance measurements were carried out. In the *dc* polarization experiments, it took 6–8 h for the polymer electrolyte samples with 0.2 mm thickness to reach steady state current, corresponding to the formation of salt concentration gradient across the film thickness. After the polarization, the complex impedance measurements were conducted again. Since the complex impedance spectra before and after polarization did not change greatly, the apparent Li^+ transference number was calculated by using the following equation [10, 11]:

$$t_{\text{Li}^+} = \frac{R_b}{\Delta V / I(\infty) - R_{\text{ct}}},$$

where ΔV is polarization voltage (10 mV), $I(\infty)$ is the steady state current after the polarization, R_b and R_{ct} are bulk resistance and charge transfer resistance, respectively, in the complex impedance spectra before the polarization. Li^+ transference number in polymer electrolytes has been the matter of argument. This method allows us to give the real Li^+ transference number if the electrolyte salts are completely dissociated and there are no mobile ion pairs [15–17]. Since ionic association takes place in such low polar media as polyethers, as has been proved by many studies [1–3], and since ion pairs in polymer electrolytes are mobile [1–3], the Li^+ transference number by this method does not give the real number but the apparent number. However, we still consider that the apparent Li^+ transference number provides a useful parameter for determining the practical merit of a polymer electrolyte in such devices as batteries, as indicated by Bruce and Vincent [15–17]. It can be noticed from Table 3 that the polymer electrolytes complexed with LiTFSI give lower t_{Li^+} values than those complexed with LiClO_4 , and that P(EO/MEEGE)-9, which forms more highly conducting polymer electrolytes, gives lower t_{Li^+} values than P(EO/MEEGE)-5 when these

polymers are complexed with the same salt. These results are exactly the same as the t_{Li^+} results of the network polymer electrolytes based on P(EO/MEEGE) [6], although the physical meaning of this behavior still remains ambiguous. We reported in the previous study [6] that the higher ionic conductivity in the LiTFSI based electrolytes than the LiClO_4 based ones originated from the higher anionic mobility of TFSI anion than that of ClO_4^- anion. We think that this consideration would be true in the present polymer electrolytes. It is interesting to note here that the t_{Li^+} values in the polymer electrolytes are much lower than those in propylene carbonate (PC) and that the difference in the polymer electrolytes between LiTFSI and LiClO_4 is opposite to the results of t_{Li^+} in PC [18, 19]. In PC, the anion (TFSI and ClO_4^-) size and the transference number show a proportionality at infinite dilution, and LiTFSI gives higher t_{Li^+} value of 0.37 than LiClO_4 ($t_{\text{Li}^+} = 0.31$). The transference number results obtained in the polymer electrolytes indicate that Li^+ ion transport in the polymer electrolytes is strongly coupled with the local segmental motion and that ClO_4^- anions interacts more strongly with the polymer segments than TFSI anion and its transport is slowed by the segmental motion.

Electrochemical stability of the polymer electrolytes has been checked by using solid state cyclic voltammetry with a microelectrode. Figure 7 shows a solid state voltammogram of P(EO/MEEGE)-5 complexed with LiTFSI at $[\text{Li}]/[\text{O}] = 0.06$, as a typical example. In the anodic scan, irreversible oxidation is seen at 1.7 V vs Ag. In the cathodic scan, lithium deposition is seen in the cathodic limit, followed by the oxidation of the Li_3Pt_2 alloy and the LiPt_2 intermetallic compound, successively, in the reverse anodic scan [20]. Consequently, the polymer electrolyte is stable up to 4.7 V from the lithium de-

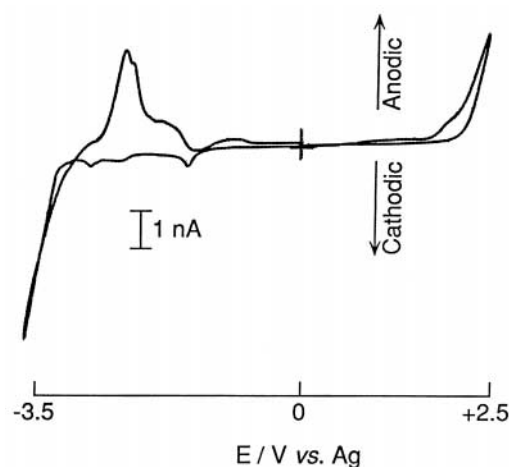


Fig. 7. Solid state cyclic voltammetry (10 mV/s) at a Pt microelectrode (10 μm diameter) for P(EO/MEEGE)-5/LiTFSI ($[\text{Li}]/[\text{O}] = 0.06$) electrolyte at 60°C.

Table 3.

Apparent Li^+ transference number for polymer electrolytes ($[\text{Li}]/[\text{O}] = 0.06$) at 60°C

Polymer	Salt	t_{Li^+}
P(EO/MEEGE)-5	LiTFSI	0.08
	LiClO_4	0.20
P(EO/MEEGE)-9	LiTFSI	0.07
	LiClO_4	0.11

position potential. The other polymer electrolytes using P(EO/MEEGE)-9 or LiClO₄ also gave at least a 4 V stable potential window from Li⁺/Li.

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